Solutions to the Exercises of Chapter 11 (Fractals)

S. H. Strogatz, *Nonlinear Dynamics and Chaos*, 2nd ed.

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Conventions. C denotes the middle-thirds Cantor set. For a bounded set S, $N(\varepsilon)$ is the minimum number of boxes of side ε needed to cover S, and the box dimension is $d = \lim_{\varepsilon \to 0} \ln N(\varepsilon) / \ln(1/\varepsilon)$. For a self-similar set made of m copies of itself, each scaled by $1/k$, the similarity dimension is $d = \ln m / \ln k$. The problem statements below are paraphrased from the book.

A counting lemma used throughout §11.3–11.4. Suppose that for $\varepsilon_n = L\,k^{-n}$ $(k > 1)$ one has $c_1 m^n \le N(\varepsilon_n) \le c_2 m^n$ for constants $c_1, c_2 > 0$. Then $d = \ln m / \ln k$. Indeed $N(\varepsilon)$ is non-increasing in ε, so for $\varepsilon_{n+1} \le \varepsilon \le \varepsilon_n$ we have $N(\varepsilon_n) \le N(\varepsilon) \le N(\varepsilon_{n+1})$, and $\ln N(\varepsilon)/\ln(1/\varepsilon) \to \ln m / \ln k$ because the constants c_i and L become negligible in the limit.

11.1 Countable and Uncountable Sets

Exercise 11.1.1

Why doesn't the diagonal argument of Example 11.1.4 show that the rationals are uncountable?

Solution. The diagonal argument builds a number y whose nth digit differs from the nth digit of the nth listed number. For the reals in $[0,1]$ this is a contradiction because y is again a real number in $[0,1]$, yet is missing from the list. For the rationals there is no contradiction: a rational number has a decimal expansion that is eventually periodic, and nothing forces the diagonal number y to be eventually periodic. In general y is irrational, so it is simply not a member of the set being listed, and no contradiction arises. (Indeed the rationals *are* countable: list p/q by increasing $|p| + q$.)

Exercise 11.1.2

Show that the set of odd integers is countable.

Solution. The map $f : \mathbb{Z} \to \{\text{odd integers}\}$, $f(n) = 2n + 1$, is a bijection (its inverse is $m \mapsto (m-1)/2$). Since $\mathbb{Z}$ is countable (list it as $0, 1, -1, 2, -2, \ldots$), so are the odd integers. Explicitly, the list is $1, -1, 3, -3, 5, -5, \ldots$

Exercise 11.1.3

Are the irrational numbers countable or uncountable? Prove it.

Solution. *Uncountable.* Suppose the irrationals I were countable. Then $\mathbb{R} = \mathbb{Q} \cup I$ would be a union of two countable sets, hence countable (interleave the two lists). But $\mathbb{R}$ is uncountable (Example 11.1.4 / Cantor's diagonal argument), a contradiction.

Exercise 11.1.4

Show, using the diagonal argument, that the set T of real numbers whose decimal expansion contains only 2's and 7's is uncountable.

Solution. Take the numbers $x \in T$ in $(0, 1)$ of the form $0.d_1d_2d_3\ldots$ with each $d_i \in \{2, 7\}$. Because the digits 0 and 9 never occur, each such number has a *unique* decimal expansion. Suppose these numbers could be listed:

$$x_1 = 0.d_{11}d_{12}d_{13}\ldots, \quad x_2 = 0.d_{21}d_{22}d_{23}\ldots, \quad \ldots$$

Define $y = 0.y_1y_2y_3\ldots$ by

$$y_n = \begin{cases} 7, & d_{nn} = 2, \\ 2, & d_{nn} = 7. \end{cases}$$

Then y has only 2's and 7's, so $y \in T$, but y differs from x_n in the nth digit and, by uniqueness of the expansion, $y \neq x_n$ for every n. So the list is incomplete, a contradiction. Hence T is uncountable.

Exercise 11.1.5

Show that the set of integer lattice points $(p, q, r) \in \mathbb{Z}^3$ is countable.

Solution. Define $g : \mathbb{Z} \to \mathbb{N} = \{0, 1, 2, \ldots\}$ by $g(p) = 2p$ for $p \geq 0$ and $g(p) = -2p - 1$ for $p < 0$; this is a bijection. Now define

$$\Phi(p, q, r) = 2^{g(p)}\, 3^{g(q)}\, 5^{g(r)} \in \mathbb{N}.$$

By uniqueness of prime factorisation, Φ is one-to-one. Thus $\mathbb{Z}^3$ is in one-to-one correspondence with an infinite subset of $\mathbb{N}$, so it is countable.

(Alternative: the points with $\max(|p|, |q|, |r|) = k$ form a finite set F_k for each $k = 0, 1, 2, \ldots$; list F_0, then F_1, then F_2, and so on.)

Exercise 11.1.6

(10x mod 1) Consider $x_{n+1} = 10x_n \pmod 1$. (a) Show that the map has countably many periodic orbits, all unstable. (b) Show that it has uncountably many aperiodic orbits. (c) Is the set of eventually-fixed points countable or uncountable?

Solution. We work on $[0, 1)$, with $f(x) = 10x - \lfloor 10x \rfloor$. Writing $x = 0.d_1d_2d_3\ldots$, the map is the shift $f(x) = 0.d_2d_3d_4\ldots$.

(a) A point has period n if $f^n(x) = x$, i.e. $10^n x - m = x$ for some integer m, i.e.

$$x = \frac{m}{10^n - 1}, \qquad m = 0, 1, \ldots, 10^n - 2.$$

So for each n there are *finitely many* points with $f^n(x) = x$. The set of all periodic points is a countable union of finite sets, hence countable, and so there are countably many periodic orbits. (Equivalently: the periodic points are exactly the numbers with purely periodic decimal expansion.) Each periodic orbit is unstable: $|f'(x)| = 10$ for all x, so $|(f^n)'(x)| = 10^n > 1$ along any orbit, i.e. the multiplier of a p-cycle is 10^p with $|10^p| > 1$.

(b) A number has an eventually periodic decimal expansion iff it is rational. Hence the orbit of every *irrational* $x \in [0, 1)$ is aperiodic (never even eventually periodic). By Exercise 11.1.3 there are uncountably many irrationals. Each orbit is itself a countable set, so uncountably many points cannot lie on only countably many orbits; therefore there are uncountably many distinct aperiodic orbits.

(c) Fixed points satisfy $10x \equiv x$, i.e. $9x \in \mathbb{Z}$, so $x = 0$ is the only fixed point in $[0, 1)$. Then x is eventually fixed iff $f^N(x) = 0$ for some N, i.e. $10^N x \in \mathbb{Z}$, i.e. $x = m/10^N$. These are the numbers with terminating decimal expansions. Each N gives finitely many such numbers, so the set of eventually-fixed points is *countable*.

Exercise 11.1.7

Show that the binary shift map $x_{n+1} = 2x_n \pmod 1$ has countably many periodic orbits and uncountably many aperiodic orbits.

Solution. The same argument as in 11.1.6 with 10 replaced by 2. Period-n points satisfy $2^n x - m = x$, i.e. $x = m/(2^n - 1)$, $m = 0, \ldots, 2^n - 2$: finitely many for each n, hence countably many periodic points and countably many periodic orbits. Rational numbers have eventually periodic binary expansions, so every irrational $x \in [0, 1)$ has an aperiodic orbit; there are uncountably many irrationals, and every orbit is countable, so there are uncountably many aperiodic orbits.

11.2 Cantor Set

Exercise 11.2.1

(Cantor set has measure zero.) By summing an infinite series, show that the total length of the removed intervals is 1.

Solution. At stage $n + 1$ ($n = 0, 1, 2, \ldots$) we remove 2^n open intervals, each of length $3^{-(n+1)}$. The total removed length is

$$\sum_{n=0}^{\infty} \frac{2^n}{3^{n+1}} = \frac{1}{3} \sum_{n=0}^{\infty} \left(\frac{2}{3}\right)^n = \frac{1}{3} \cdot \frac{1}{1 - 2/3} = 1.$$

The unit interval has length 1, so the leftover set C has length $1 - 1 = 0$.

Exercise 11.2.2

Show that the rationals have zero measure.

Solution. The rationals are countable, so list them as $q_1, q_2, q_3, \ldots$. Fix $\epsilon > 0$ and cover q_n by an open interval of length $\epsilon/2^n$. These intervals cover $\mathbb{Q}$ and their total length is at most

$$\sum_{n=1}^{\infty} \frac{\epsilon}{2^n} = \epsilon.$$

Since ϵ is arbitrary, the measure of $\mathbb{Q}$ is zero.

Exercise 11.2.3

Show that any countable subset of $\mathbb{R}$ has zero measure.

Solution. Let $A = \{a_1, a_2, \ldots\}$ be countable (finite sets are trivial). The proof of 11.2.2 used nothing about the rationals except the existence of the list: cover a_n by an interval of length $\epsilon/2^n$; the total length is at most ϵ for every $\epsilon > 0$, so A has measure zero.

Exercise 11.2.4

Consider the set J of irrational numbers between 0 and 1. (a) What is its measure? (b) Countable or uncountable? (c) Totally disconnected? (d) Any isolated points?

Solution.

(a) $[0, 1] = (\mathbb{Q} \cap [0, 1]) \cup J$ is a disjoint union, and the rationals have measure 0 (11.2.2). So J has measure $1 - 0 = 1$.

(b) Uncountable (Exercise 11.1.3; the same proof works inside $(0, 1)$).

(c) Yes. J contains no interval: between any two numbers there is a rational, which is not in J. In fact, any two irrationals can be separated by a rational number, so J is totally disconnected.

(d) No. Every neighbourhood of an irrational number x contains other irrationals (e.g. $x+\sqrt{2}/n$ for large n, as long as it stays in $(0,1)$). So J has no isolated points.

(Thus J shares the topological properties of the Cantor set, yet has measure 1 rather than 0.)

Exercise 11.2.5

(Base 3 and the Cantor set.) (a) Find the base-3 expansion of $1/2$. *(b) Find a one-to-one correspondence between* C *and* $[0,1]$. *(c) Show that not every point of* C *is an endpoint of a construction interval.*

Solution. **(a)** Since $\sum_{k\geq 1} 3^{-k} = \frac{1/3}{1-1/3} = \frac{1}{2}$, we have $\frac{1}{2} = 0.1111\ldots_3$.

(b) *Description of* C *in base 3.* The first stage removes $(1/3, 2/3)$, i.e. the numbers whose first ternary digit must be 1; the next stage removes numbers whose second digit must be 1; and so on. Hence C consists exactly of the numbers in $[0,1]$ that have *some* ternary expansion using only the digits 0 and 2. (Endpoints have two expansions; e.g. $\frac{1}{3} = 0.1000\ldots_3 = 0.0222\ldots_3$, and the second one only uses $0, 2$.)

The map. For $c = \sum_{k\geq 1} a_k 3^{-k} \in C$ with $a_k \in \{0,2\}$ define

$$\varphi(c) = \sum_{k\geq 1} \frac{a_k}{2}\, 2^{-k} \in [0,1],$$

i.e. replace every 2 by 1 and read the result in base 2. Conversely, for $x = \sum b_k 2^{-k} \in [0,1]$, $b_k \in \{0,1\}$, the inverse rule is $\psi(x) = \sum 2b_k 3^{-k} \in C$. Then $\varphi \circ \psi = \mathrm{id}$, so φ is onto $[0,1]$, and φ is one-to-one except for the countably many pairs of points that are the two endpoints of a removed middle-third gap (e.g. $\varphi(1/3) = \varphi(0.0222\ldots_3) = 0.0111\ldots_2 = \frac{1}{2} = \varphi(2/3) = \varphi(0.2000\ldots_3)$), which correspond to the two binary expansions of a dyadic rational.

An exact bijection. Fixing the non-terminating binary expansion for each $x \in (0,1]$ makes $\psi : [0,1] \to C$ one-to-one; the inclusion $C \hookrightarrow [0,1]$ is also one-to-one. By the Cantor–Schröder–Bernstein theorem there is a genuine bijection $C \leftrightarrow [0,1]$ (it differs from φ only on a countable set). In particular C is uncountable, with "as many points" as $[0,1]$.

(c) Take $x = \frac{1}{4}$. Summing a geometric series,

$$\sum_{k\geq 1} 2\cdot 3^{-2k} = \frac{2/9}{1-1/9} = \frac{1}{4}, \qquad \text{so} \qquad \tfrac{1}{4} = 0.020202\ldots_3\,.$$

Its expansion uses only 0 and 2, so $\frac{1}{4} \in C$. The endpoints of the construction intervals are the numbers whose ternary expansion is a finite string of $0, 2$ followed by all 0's or all 2's (e.g. $0.0222\ldots_3$, $0.2000\ldots_3$). Since $\frac{1}{4}$ is not of the form $m/3^n$, its ternary expansion is unique and is not eventually constant, so $\frac{1}{4}$ is *not* an endpoint. In fact C has uncountably many non-endpoints, because the set of endpoints is countable while C is not.

Exercise 11.2.6

(Devil's staircase.) Let $P(x)$ *be the probability that a randomly chosen point of the Cantor set lies to the left of* x. *(a) Show* $P_0(x) = x$. *(b) Graph* P_1. *(c) Graph* P_2, P_3, P_4. *(d) Is the limit* P_d *continuous? What does its derivative look like?*

Solution. **(a)** $S_0 = [0,1]$, and a point chosen uniformly at random in $[0,1]$ lies to the left of x with probability $P_0(x) = x$.

(b) $S_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$. Each piece carries probability $\frac{1}{2}$, spread uniformly over length $\frac{1}{3}$, i.e. with density $3/2$. Hence

$$P_1(x) = \begin{cases} \frac{3}{2}x, & 0 \leq x \leq \frac{1}{3}, \\ \frac{1}{2}, & \frac{1}{3} \leq x \leq \frac{2}{3} \text{ (plateau)}, \\ \frac{1}{2} + \frac{3}{2}\big(x - \frac{2}{3}\big), & \frac{2}{3} \leq x \leq 1. \end{cases}$$

(c) S_n consists of 2^n intervals of length 3^{-n}, each carrying probability 2^{-n}. So P_n is piecewise linear: it rises with slope $(3/2)^n$ across each interval of S_n and is constant (a plateau) across each

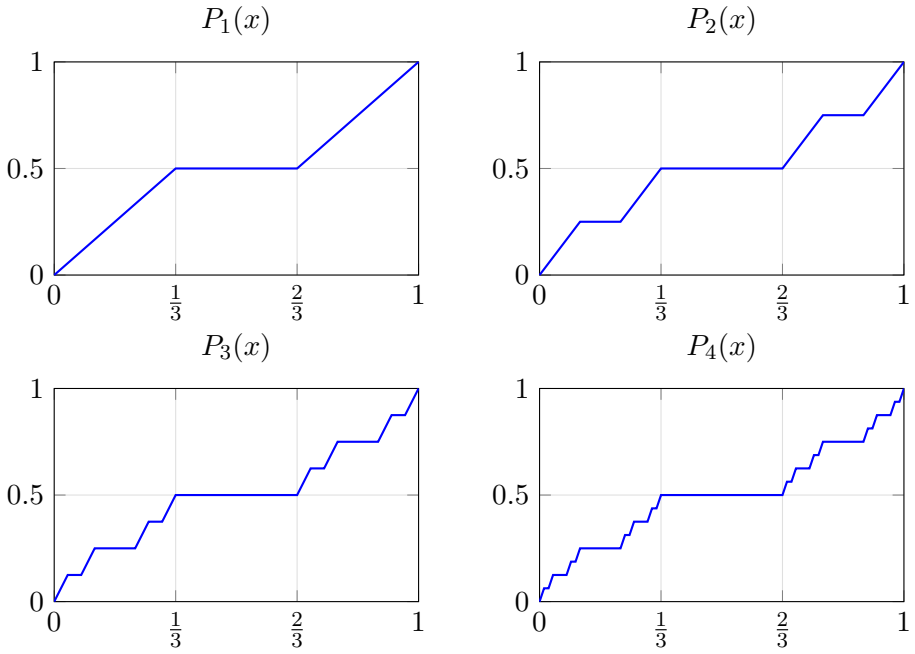


Figure 1: The approximants $P_1, \ldots, P_4$ to the devil's staircase.

of the removed gaps. For example, P_2 has slope $9/4$ on $[0, \frac{1}{9}]$ (rising from 0 to $\frac{1}{4}$), plateau value $\frac{1}{4}$ on $[\frac{1}{9}, \frac{2}{9}]$, slope $\frac{9}{4}$ on $[\frac{2}{9}, \frac{1}{3}]$ (rising from $\frac{1}{4}$ to $\frac{1}{2}$), plateau $\frac{1}{2}$ on $[\frac{1}{3}, \frac{2}{3}]$, and symmetrically on $[\frac{2}{3}, 1]$. The plateau heights are always multiples of 2^{-n}, and each stage $n \to n+1$ replaces every sloping piece by "slope–plateau–slope".

(d) All P_n are continuous and non-decreasing, and for $m \geq n$ the functions P_n and P_m agree at all endpoints of S_n-intervals (where both equal an integer multiple of 2^{-n}) and are monotone in between, so $\sup_x |P_m(x) - P_n(x)| \leq 2^{-n}$. Hence $P_n \to P_d$ *uniformly*, and the uniform limit of continuous functions is continuous. Thus the devil's staircase P_d is continuous and non-decreasing, with $P_d(0) = 0$, $P_d(1) = 1$. On each removed gap it is constant, so $P_d'(x) = 0$ on the complement of the Cantor set, which has total length 1. So the graph of P_d' is zero "almost everywhere", yet $\int_0^1 P_d' \, dx = 0 \neq P_d(1) - P_d(0) = 1$: all of the rise of P_d happens on the measure-zero set C, where P_d' is infinite or undefined (it behaves like a density concentrated on C, not an ordinary function). So P_d is not the integral of its derivative.

11.3 Dimension of Self-Similar Fractals

Exercise 11.3.1

(Middle-halves Cantor set.) Remove the middle half of each sub-interval. (a) Find the similarity dimension. (b) Find the measure.

Solution. **(a)** Removing the middle half of $[0, 1]$ leaves $[0, \frac{1}{4}] \cup [\frac{3}{4}, 1]$: $m = 2$ copies, each scaled by $1/k = \frac{1}{4}$. So

$$d = \frac{\ln 2}{\ln 4} = \frac{1}{2}.$$

(b) After n stages the remaining length is $(1/2)^n \to 0$ (each stage keeps half the length), so the set has measure zero.

Exercise 11.3.2

(Generalised Cantor set.) Remove an open middle interval of length fraction $0 < a < 1$ at each stage. Find the similarity dimension.

Solution. Each stage replaces an interval of length ℓ by two intervals of length $\frac{1-a}{2}\ell$. Hence $m = 2$ copies at scale $1/k = (1-a)/2$:

$$d = \frac{\ln 2}{\ln\left(\frac{2}{1-a}\right)}.$$

Checks: $a = \frac{1}{3}$ gives $\ln 2/\ln 3$; $a = \frac{1}{2}$ gives $\ln 2/\ln 4 = \frac{1}{2}$ (11.3.1). As $a \to 0$, $d \to 1$, and as $a \to 1$, $d \to 0$.

Exercise 11.3.3

(Even-sevenths Cantor set.) Divide $[0,1]$ *into seven equal pieces, delete pieces 2, 4, 6, and repeat. (a) Find the similarity dimension. (b) Generalise to any odd number of pieces.*

Solution. **(a)** Pieces $1, 3, 5, 7$ remain: $m = 4$ copies, each scaled by $\frac{1}{7}$:

$$d = \frac{\ln 4}{\ln 7} \approx 0.712.$$

(b) Divide into $n = 2j + 1$ pieces and delete the j even-numbered ones. The $j + 1$ odd-numbered pieces remain, each scaled by $1/n$:

$$d = \frac{\ln(j+1)}{\ln(2j+1)} = \frac{\ln \frac{n+1}{2}}{\ln n}.$$

For $n = 3$ this is the middle-thirds Cantor set, $\ln 2/\ln 3$; for $n = 7$ it is $\ln 4/\ln 7$.

Exercise 11.3.4

(No odd digits.) Find the similarity dimension of the set of numbers in $[0,1]$ *whose decimal expansions have only even digits.*

Solution. The first digit must be one of $0, 2, 4, 6, 8$: the set is the union of $m = 5$ copies of itself, each scaled by $\frac{1}{10}$ (the first digit fixes a subinterval of length $\frac{1}{10}$, and the remaining digits again have only even digits). So

$$d = \frac{\ln 5}{\ln 10} = \log_{10} 5 \approx 0.699.$$

Exercise 11.3.5

(No 8's.) Find the similarity dimension of the numbers in $[0,1]$ *with no digit 8 in their decimal expansion.*

Solution. The first digit can be any of the nine digits $0, 1, \ldots, 7, 9$, and after that the same rule applies, so the set is $m = 9$ copies of itself scaled by $\frac{1}{10}$:

$$d = \frac{\ln 9}{\ln 10} \approx 0.954.$$

Exercise 11.3.6

Show that C *contains no intervals, but that no point of* C *is isolated.*

Solution. *No intervals.* After n stages C is contained in 2^n closed intervals of length 3^{-n}. An interval of length $\delta > 0$ contained in C would have to fit inside one of these for every n, which is impossible once $3^{-n} < \delta$.

No isolated points. Let $x \in C$ and $\eta > 0$. Choose n with $3^{-n} < \eta$. Then x lies in one of the stage-n intervals I, of length 3^{-n}. The two endpoints of I belong to C (an endpoint is never removed in later stages, since only open middle thirds are removed), and at least one of them differs from x. That endpoint is in C and within $3^{-n} < \eta$ of x. So every neighbourhood of x contains another point of C.

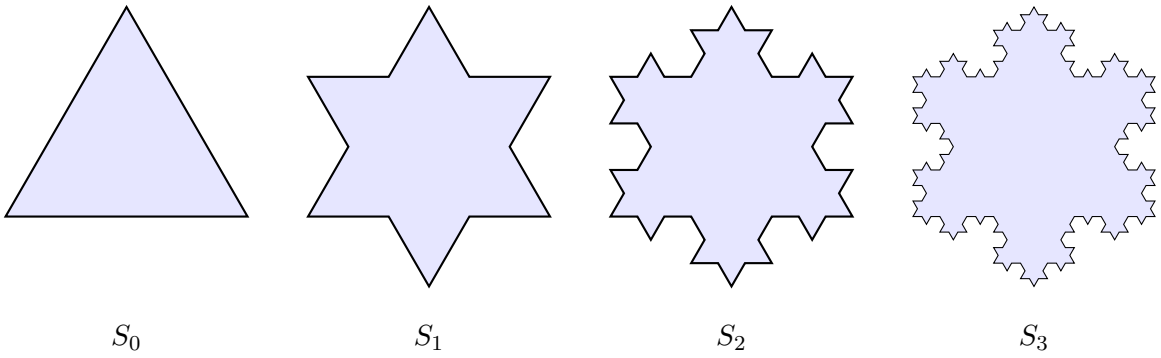


Figure 2: S_0, S_1, S_2, S_3 for the von Koch snowflake.

Exercise 11.3.7
(Snowflake.) Start with an equilateral triangle S_0 and apply the von Koch procedure to each side. (a) Show S_1 is a star of David. (b) Draw S_2, S_3. (c) Show the limiting curve has infinite length. (d) Find the enclosed area. (e) Find the similarity dimension.

Solution. Let the triangle have side $s = 1$.

(a) Cut off the three corners of the triangle, each a small equilateral triangle of side $\frac{1}{3}$. What remains is a hexagon all of whose angles are 120° and whose six sides all have length $\frac{1}{3}$, i.e. a regular hexagon. Three of its sides are the middle thirds of the original sides; the other three are the cuts. The von Koch step puts an equilateral triangle of side $\frac{1}{3}$ on each middle third. So S_1 = hexagon $\cup$ three corner triangles $\cup$ three new bumps = a regular hexagon with an equilateral triangle of side $\frac{1}{3}$ attached to each of its six sides. That is the six-pointed star of David.

(b) See Figure 2.
(c) S_n has $3 \cdot 4^n$ sides, each of length 3^{-n} (every step replaces each side by 4 sides one-third as long). Its perimeter is

$$L_n = 3\Big(\frac{4}{3}\Big)^n \xrightarrow[n\to\infty]{} \infty.$$

All vertices of S_n lie on the limiting curve S (a vertex is never removed later), so S_n is a polygon inscribed in S and $\text{length}(S) \geq L_n$ for every n. Hence S has infinite arc length.

(d) Let $A_0 = \frac{\sqrt{3}}{4}$ be the area of S_0. The step $S_{n-1} \to S_n$ adds one small triangle per side of S_{n-1}, i.e. $3 \cdot 4^{n-1}$ triangles of side 3^{-n}, each of area $A_0/9^n$. Therefore

$$A(S) = A_0\Big[1 + \sum_{n=1}^{\infty} \frac{3 \cdot 4^{n-1}}{9^n}\Big] = A_0\Big[1 + \frac{1}{3}\sum_{j=0}^{\infty}\Big(\frac{4}{9}\Big)^j\Big] = A_0\Big[1 + \frac{1}{3} \cdot \frac{9}{5}\Big] = \frac{8}{5}A_0 = \frac{2\sqrt{3}}{5} \approx 0.693.$$

(For a general side length s, multiply by s^2.) So the area is $\frac{8}{5}$ that of the original triangle: finite, even though the perimeter is infinite.

(e) Each side is replaced by $m = 4$ copies scaled by $\frac{1}{3}$, so

$$d = \frac{\ln 4}{\ln 3} \approx 1.262.$$

Exercise 11.3.8
(Sierpinski carpet.) Divide the closed unit square into nine equal squares and delete the open central one; repeat. (a) Sketch S_3. (b) Find the similarity dimension. (c) Show the carpet has zero area.

Solution. **(a)** See Figure 3.
(b) The carpet consists of $m = 8$ copies of itself, each scaled by $\frac{1}{3}$:

$$d = \frac{\ln 8}{\ln 3} = \frac{3\ln 2}{\ln 3} \approx 1.893.$$

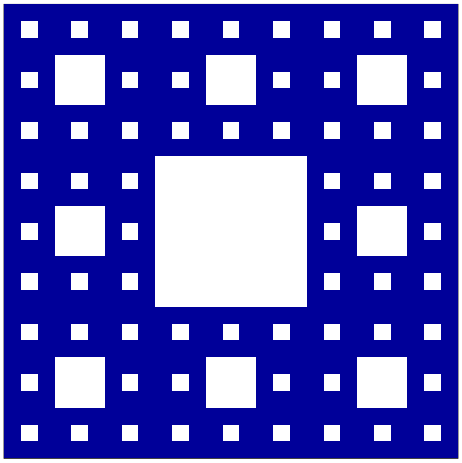


Figure 3: The stage S_3 of the Sierpinski carpet ($8^3 = 512$ squares of side $1/27$).

(c) Each stage keeps $\frac{8}{9}$ of the area, so $\text{Area}(S_n) = (8/9)^n \to 0$. The carpet is contained in every S_n, so its area is zero.

Exercise 11.3.9

(Sponges.) Divide a solid cube into 27 sub-cubes and delete the central cube of each face together with the central cube of the cube; iterate (Menger sponge). Find the similarity dimension, and generalise to N dimensions.

Solution. 3-D: of the 27 sub-cubes, 6 face-centre cubes and 1 body-centre cube are removed, leaving $m = 20$ (8 corners and 12 edges). Each copy is scaled by $\frac{1}{3}$:

$$d = \frac{\ln 20}{\ln 3} \approx 2.727.$$

N dimensions. Divide the N-cube into 3^N sub-cubes, labelled by $(i_1, \dots, i_N) \in \{0,1,2\}^N$ (where the digit 1 means "middle"). In $N = 3$, the removed cubes are exactly those having *at least two* middle coordinates (a face centre has two, the body centre has three); the kept ones have at most one. Taking this as the definition for general N (for $N = 2$ it gives the Sierpinski carpet, for $N = 3$ the Menger sponge), the number of kept sub-cubes is

$$m_N = \underbrace{2^N}_{\text{no middle coordinate}} + \underbrace{N\,2^{N-1}}_{\text{exactly one}} = 2^{N-1}(N+2),$$

(check: $m_2 = 8$, $m_3 = 20$), each scaled by $\frac{1}{3}$, so

$$d_N = \frac{\ln\bigl(2^{N-1}(N+2)\bigr)}{\ln 3}.$$

For instance $d_2 = \ln 8/\ln 3$, $d_3 = \ln 20/\ln 3$, $d_4 = \ln 48/\ln 3 \approx 3.524$.

Exercise 11.3.10

(Fat fractal.) Start with $[0,1]$ and delete the open middle $1/2, 1/4, 1/8, \dots$ of each remaining sub-interval (at stage n, a fraction 2^{-n}). (a) Show the limit set is a topological Cantor set. (b) Show its measure is positive; find it or a lower bound.

Solution. **(a)** A topological Cantor set is a closed set that contains no intervals and has no isolated points.

Closed: it is an intersection of finite unions of closed intervals.

No intervals: at stage n there are 2^n intervals; each is obtained from an interval of length ℓ at the previous stage by removing the open middle fraction $2^{-n} \le \frac{1}{2}$ and keeping two pieces of length $\frac{1}{2}(1 - 2^{-n})\ell \le \frac{1}{2}\ell$. So all stage-$n$ intervals have length at most 2^{-n}, which tends to 0. An interval of positive length cannot lie in the set.

No isolated points: as in 11.3.6, if x is in the set, then it lies in a stage-n interval of length $\le 2^{-n}$ whose endpoints are never removed. One of the endpoints differs from x and is within 2^{-n} of it.

(b) At stage n the remaining total length is multiplied by $1-2^{-n}$ (we remove a fraction 2^{-n} of what is left). Hence the measure of the limit set is

$$\mu = \prod_{n=1}^{\infty}\bigl(1-2^{-n}\bigr) = \frac{1}{2}\cdot\frac{3}{4}\cdot\frac{7}{8}\cdot\frac{15}{16}\cdots \approx 0.2888.$$

A simple lower bound: using $\prod(1-a_n) \ge 1-\sum a_n$ for $a_n \in [0,1]$, apply it to the factors with $n \ge 2$:

$$\mu \ge \frac{1}{2}\Bigl(1-\sum_{n=2}^{\infty}2^{-n}\Bigr) = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4} > 0.$$

The numerical value $0.28879\ldots$ is the value of the Euler function $\phi(\tfrac12) = \prod(1-2^{-n})$. So the set has positive measure, which is why it's called a "fat" fractal.

11.4 Box Dimension

Exercise 11.4.1

Find the box dimension of the von Koch snowflake.

Solution. Take $s = 1$ and boxes of side $\varepsilon_n = 3^{-n}$. The snowflake curve is contained in the union of the $3\cdot 4^n$ segments of S_n, each of length 3^{-n}, and each such segment meets at most 3 boxes of a grid of side 3^{-n} (up to a constant), so $N(\varepsilon_n) \le c_2\, 4^n$. Conversely the $3\cdot 4^n$ vertices of S_n lie on the curve and any two of them are at least 3^{-n} apart, so a box of side 3^{-n} contains at most a bounded number of them, giving $N(\varepsilon_n) \ge c_1 4^n$. By the counting lemma,

$$d_{\text{box}} = \frac{\ln 4}{\ln 3} \approx 1.262,$$

the same as the similarity dimension.

Exercise 11.4.2

Find the box dimension of the Sierpinski carpet.

Solution. At stage n the carpet lies in 8^n squares of side 3^{-n} (and each of these squares contains points of the carpet, e.g. its corners). So with $\varepsilon_n = 3^{-n}$ we get $N(\varepsilon_n) = 8^n$, and

$$d_{\text{box}} = \frac{\ln 8}{\ln 3} \approx 1.893.$$

Exercise 11.4.3

Find the box dimension of the Menger sponge.

Solution. At stage n the sponge lies in 20^n cubes of side 3^{-n}, each containing points of the sponge, so $N(3^{-n}) = 20^n$ and

$$d_{\text{box}} = \frac{\ln 20}{\ln 3} \approx 2.727.$$

Exercise 11.4.4

Find the box dimension of $C \times C$.

Solution. C at stage n consists of 2^n intervals of length 3^{-n}. Their product covers $C \times C$ with $2^n \cdot 2^n = 4^n$ squares of side 3^{-n} (each containing points of $C \times C$). Thus $N(3^{-n}) = 4^n$ and

$$d_{\text{box}} = \frac{\ln 4}{\ln 3} = 2\,\frac{\ln 2}{\ln 3} \approx 1.262,$$

twice the dimension of C, as expected for a product.

Exercise 11.4.5

Find the box dimension of the Menger hypersponge (Exercise 11.3.9).

Solution. Using the N-dimensional construction of 11.3.9, the stage-n set consists of $m_N^n = \big(2^{N-1}(N+2)\big)^n$ cubes of side 3^{-n}, so

$$d_{\text{box}} = \frac{\ln\big(2^{N-1}(N+2)\big)}{\ln 3}.$$

For $N = 2, 3$ this reproduces 11.4.2 and 11.4.3.

Exercise 11.4.6

(A strange repeller for the tent map.) The tent map $f(x) = rx$ for $0 \le x \le \frac{1}{2}$, $f(x) = r(1-x)$ for $\frac{1}{2} \le x \le 1$, with $r > 2$. (a) Find the points that escape after one or two iterations. (b) Describe the set that never escapes. (c) Find its box dimension. (d) Show that the Liapunov exponent is positive on it.

Solution. Since $r > 2$, we have $1/r < 1/2 < 1 - 1/r$.

(a) *One iteration.* $f(x) > 1$ iff $rx > 1$ (left branch, $x > 1/r$) or $r(1-x) > 1$ (right branch, $x < 1 - 1/r$). So the points that escape after exactly one iteration are

$$E_1 = \Big(\frac{1}{r},\ 1 - \frac{1}{r}\Big),$$

an open interval around $\frac{1}{2}$ (the peak of the tent).

Two iterations. We need $f(x) \in [0,1]$ but $f(x) \in E_1$, i.e. $f(x) \in (\frac{1}{r}, 1 - \frac{1}{r})$. On the left branch, $rx \in (\frac{1}{r}, 1 - \frac{1}{r})$ gives $x \in (\frac{1}{r^2}, \frac{1}{r} - \frac{1}{r^2})$. On the right branch, $r(1-x) \in (\frac{1}{r}, 1 - \frac{1}{r})$ gives $x \in (1 - \frac{1}{r} + \frac{1}{r^2},\ 1 - \frac{1}{r^2})$. So

$$E_2 = \Big(\frac{1}{r^2}, \frac{1}{r} - \frac{1}{r^2}\Big) \cup \Big(1 - \frac{1}{r} + \frac{1}{r^2},\ 1 - \frac{1}{r^2}\Big),$$

one open interval in the middle of each of the two intervals $[0, \frac{1}{r}]$ and $[1 - \frac{1}{r}, 1]$ that remain after the first step.

(b) Let $\Lambda_n = \{x : f^k(x) \in [0,1],\ k = 0, \dots, n\}$. Then $\Lambda_0 = [0,1]$, $\Lambda_1 = [0, \frac{1}{r}] \cup [1 - \frac{1}{r}, 1]$, and in general $\Lambda_n = \{x \in [0,1] : f(x) \in \Lambda_{n-1}\}$. Each branch of f is linear with slope $\pm r$ and maps onto a set containing $[0,1]$, so by induction Λ_n consists of 2^n closed intervals of length r^{-n}, and each is obtained from an interval of Λ_{n-1} by removing an open interval from the *middle*, namely the fraction $1 - \frac{2}{r}$ of its length. The set of points that never escape is

$$\Lambda = \bigcap_{n \ge 0} \Lambda_n,$$

which is a Cantor set: closed, without intervals (lengths $r^{-n} \to 0$), and without isolated points (by the endpoint argument of 11.3.6). For $r = 3$ it is exactly the middle-thirds Cantor set.

(c) Using boxes of side $\varepsilon_n = r^{-n}$: $N(\varepsilon_n) = 2^n$. Hence

$$d_{\text{box}} = \frac{\ln 2}{\ln r}, \qquad r > 2.$$

($r = 3$ gives $\ln 2 / \ln 3$; $d \to 1$ as $r \to 2^+$, and $d \to 0$ as $r \to \infty$.)

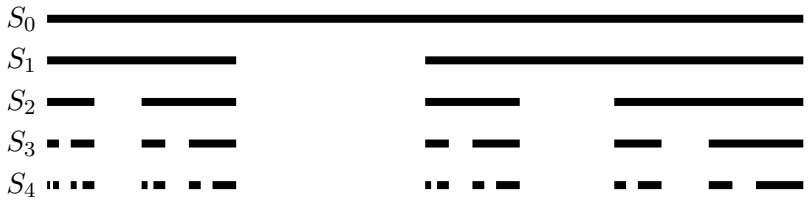


Figure 4: $S_0, \dots, S_4$ for the lopsided fractal (the numbers of intervals are $1, 2, 4, 8, 16$, with varying lengths).

(d) For $x \in \Lambda$, every iterate $x_k = f^k(x)$ stays in $\Lambda \subset [0, \frac{1}{r}] \cup [1 - \frac{1}{r}, 1]$, which excludes the corner $x = \frac{1}{2}$. Therefore $|f'(x_k)| = r$ for all k, and

$$\lambda = \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \ln |f'(x_k)| = \ln r > \ln 2 > 0.$$

So nearby orbits separate exponentially, and Λ is chaotic (a "strange repeller").

Exercise 11.4.7

(A lopsided fractal.) Divide $[0, 1]$ *into four quarters and delete the open second quarter from the left, giving* S_1*. Generate* S_{n+1} *from* S_n *by deleting the second quarter of each of the intervals of* S_n*. (a) Sketch* $S_1, \dots, S_4$*. (b) Find the box dimension of* S_∞*. (c) Is* S_∞ *self-similar?*

Solution. **(a)** The key observation is that deleting the second quarter leaves the first quarter and the union of the 3rd and 4th quarters, which is *one* interval of length $\frac{1}{2}$: $S_1 = [0, \frac{1}{4}] \cup [\frac{1}{2}, 1]$. Then, treating each interval of S_1 separately, $S_2 = [0, \frac{1}{16}] \cup [\frac{1}{8}, \frac{1}{4}] \cup [\frac{1}{2}, \frac{5}{8}] \cup [\frac{3}{4}, 1]$, and so on. A sketch is given below.

(b) Each interval I of length ℓ in S_n is replaced by two intervals, of lengths $\ell/4$ and $\ell/2$. Since the two parts of S_1 are separated by a gap, the construction acts on them independently, so S_∞ is the union of two scaled copies of itself:

$$S_\infty = \tfrac{1}{4} S_\infty \ \cup \ \left(\tfrac{1}{2} S_\infty + \tfrac{1}{2}\right).$$

The scale factors are $\frac{1}{4}$ and $\frac{1}{2}$ (not equal), and the dimension d satisfies the Moran equation

$$\left(\frac{1}{4}\right)^d + \left(\frac{1}{2}\right)^d = 1.$$

With $u = 2^{-d}$ this is $u^2 + u - 1 = 0$, so $u = \frac{\sqrt{5}-1}{2} = \frac{1}{\varphi}$, where $\varphi = \frac{1+\sqrt{5}}{2}$ is the golden ratio. Therefore

$$d_{\text{box}} = \frac{\ln \varphi}{\ln 2} = \frac{\ln \frac{1+\sqrt{5}}{2}}{\ln 2} \approx 0.694.$$

(The two pieces are disjoint and separated by a gap, so the open set condition holds and the box dimension equals the similarity dimension given by this equation. Covering by the 2^n intervals of stage n, whose lengths are $2^{-a}4^{-b}$ with $a + b = n$, gives the upper bound.)

(c) Yes, S_∞ is self-similar, but not with a single scale factor: it is the union of two scaled copies of itself, one reduced by $\frac{1}{4}$ and one by $\frac{1}{2}$. Because the ratios differ, the dimension comes from the Moran equation above rather than from $\ln m / \ln k$. (The naive guess $\ln 3 / \ln 4 \approx 0.79$, which assumes three pieces of size $\frac{1}{4}$, is wrong because the 3rd and 4th quarters merge into one interval.)

Exercise 11.4.8

(Random fractals.) As in 11.4.7, but flip a coin to generate S_{n+1} *from* S_n*: heads* $\Rightarrow$ *delete the second quarter of every interval, tails* $\Rightarrow$ *delete the third quarter. (a) Can you find the box dimension? Does the question make sense? (b) What if tails means delete the* first *quarter?*

Solution. **(a)** Deleting the second quarter of an interval of length ℓ leaves pieces of lengths $\ell/4$ and $\ell/2$ (in this order); deleting the third quarter leaves pieces of lengths $\ell/2$ and $\ell/4$ (the mirror image). So whichever way the coin falls, every interval of length ℓ is replaced by two intervals of lengths $\ell/4$ and $\ell/2$. Hence at every stage, the *multiset of interval lengths* is the same for all sequences of heads and tails: after n steps there are $\binom{n}{j}$ intervals of length 2^{-n-j}, $j = 0, \dots, n$. Only the *positions* of the intervals depend on the coin flips. The box-counting dimension is determined by how the lengths shrink, and the quantity

$$\sum_i \ell_i^d = \left(4^{-d} + 2^{-d}\right)^n$$

is independent of the flip sequence. Exactly as in 11.4.7, this singles out $4^{-d} + 2^{-d} = 1$, so, for *every* sequence of flips,

$$d_{\text{box}} = \frac{\ln \varphi}{\ln 2} \approx 0.694.$$

So the question makes sense, and the answer does not depend on the particular sequence of heads and tails This can be made rigorous by the usual Moran-covering argument, since at every stage the gap between the two pieces is a fixed fraction of the parent interval (see Falconer, Ch. 15).

(b) Now tails removes the first quarter and leaves a *single* interval of length $\frac{3}{4}\ell$, with no splitting. Heads (second quarter deleted) still replaces ℓ by $\ell/4, \ell/2$. This does make a difference. A long run of tails shrinks every interval by $\frac{3}{4}$ per step without creating any new intervals: e.g. for all tails, $S_n = [1 - (3/4)^n, 1] \to \{1\}$, a single point, with dimension 0. Hence the dimension *depends on the sequence*: the sum $\sum \ell_i^d$ is multiplied by $4^{-d} + 2^{-d}$ at a head and by $(3/4)^d$ at a tail. If heads occur with frequency $\frac{1}{2}$, then one expects (heuristically, by the law of large numbers) the dimension d solving

$$\left(4^{-d} + 2^{-d}\right)\left(\tfrac{3}{4}\right)^d = 1 \quad \Longrightarrow \quad d \approx 0.535,$$

for typical sequences, which is smaller than 0.694. But a different sequence (e.g. with long strings of tails) gives a different value, and for badly behaved sequences the box dimension may fail to exist because the scaling differs at different scales.

Exercise 11.4.9

(Fractal cheese.) Divide the unit square into p^2 squares, choose m^2 of them (at random) and discard them, where $p > m \geq 1$ are integers. Repeat for each remaining square. Find the box dimension.

Solution. After the first stage there remain $p^2 - m^2$ squares of side $1/p$; each is treated in the same way, so after n stages there are exactly $(p^2 - m^2)^n$ squares of side p^{-n}, *whatever* choice of discarded squares was made at each step, and each of them contains points of the limiting set (e.g. its corners, which are never deleted if the discarded squares are open). With $\varepsilon_n = p^{-n}$ we thus have $N(\varepsilon_n) = (p^2 - m^2)^n$ and

$$d_{\text{box}} = \frac{\ln(p^2 - m^2)}{\ln p}.$$

(For $p = 3$, $m = 1$ this is the Sierpinski carpet, $\ln 8 / \ln 3$.) The set need not be self-similar, but the count is the same.

Exercise 11.4.10

(Fat fractal.) Show that the fat fractal of 11.3.10 has box dimension 1.

Solution. Let F be the fat fractal, with measure $\mu(F) > 0$ (11.3.10). Since $F \subset [0,1]$, $N(\varepsilon) \leq 1/\varepsilon + 1$, so $d_{\text{box}} \leq 1$. Conversely, $N(\varepsilon)$ boxes of side ε that cover F have total length $N(\varepsilon)\varepsilon \geq \mu(F)$, so $N(\varepsilon) \geq \mu(F)/\varepsilon$ and

$$d_{\text{box}} = \lim_{\varepsilon \to 0} \frac{\ln N(\varepsilon)}{\ln(1/\varepsilon)} \geq \lim_{\varepsilon \to 0} \frac{\ln \mu(F) + \ln(1/\varepsilon)}{\ln(1/\varepsilon)} = 1.$$

Hence $d_{\text{box}} = 1$, even though F is a Cantor set.

11.5 Pointwise and Correlation Dimensions

Exercise 11.5.1

(Project.) Write a program to compute the correlation dimension of the Lorenz attractor. Reproduce the results of Figure 11.5.3. Then try other values of r. How does the dimension depend on r?

Solution. Method (Grassberger–Procaccia). Integrate the Lorenz equations $\dot{x} = \sigma(y - x)$, $\dot{y} = rx - y - xz$, $\dot{z} = xy - bz$ with $\sigma = 10$, $b = 8/3$. Discard a transient, then record $n = 15{,}000$ points $\mathbf{x}_i$ at time spacing $\Delta t = 0.25$ (the same numbers as in the book's example). The correlation integral is

$$C(\varepsilon) = \frac{\#\{(i,j) : i \neq j,\ \|\mathbf{x}_i - \mathbf{x}_j\| < \varepsilon\}}{n(n-1)} \ \sim\ \varepsilon^{d},$$

and d is the slope of $\log C$ against $\log \varepsilon$ in the scaling region (small compared to the attractor's diameter, large compared to the point spacing). The code is below (Python, using SciPy's ODE solver and a k-d tree to count pairs efficiently).

```
import numpy as np
from scipy.integrate import solve_ivp
from scipy.spatial import cKDTree

def lorenz(t, u, sigma=10.0, b=8.0/3.0, r=28.0):
    x, y, z = u
    return [sigma*(y-x), r*x - y - x*z, x*y - b*z]

def attractor_points(r, n=15000, dt=0.25, t_transient=100.0, seed=0):
    rng = np.random.default_rng(seed)
    u0 = np.array([1.0, 1.0, r-1.0]) + 0.1*rng.standard_normal(3)
    s = solve_ivp(lorenz, [0, t_transient], u0, args=(10.0, 8/3, r),
                  rtol=1e-10, atol=1e-12, method="DOP853")
    t = dt*np.arange(n)
    s = solve_ivp(lorenz, [0, t[-1]], s.y[:, -1], t_eval=t,
                  args=(10.0, 8/3, r), rtol=1e-10, atol=1e-12,
                  method="DOP853")
    return s.y.T

def correlation_integral(P, eps):
    tree = cKDTree(P)
    cnt = tree.count_neighbors(tree, eps) - len(P)   # remove i=j pairs
    return cnt / (len(P)*(len(P)-1))

def fit_dimension(P, lo=0.005, hi=0.1, k=25):
    diam = np.ptp(P, axis=0).max()
    eps = np.geomspace(lo*diam, hi*diam, k)          # scaling region
    C = correlation_integral(P, eps)
    ok = C > 0
    return np.polyfit(np.log(eps[ok]), np.log(C[ok]), 1)[0]

for r in [25, 28, 35, 45, 60, 100, 150, 166]:
    print(r, fit_dimension(attractor_points(r)))
```

Result at $r = 28$. Figure 5 shows $\log_2 C(\varepsilon)$ against $\log_2(1/\varepsilon)$. It has the same shape as Figure 11.5.3: a straight scaling region, saturation at large ε (where the balls engulf the attractor, $C \to 1$), and the fitted slope

$$d_{\mathrm{corr}} \approx 2.07,$$

in agreement with the book's value 2.05 ± 0.01 (the small difference is due to the choice of fit window and the finite sample of 15,000 points).

Dependence on r. Results from the same program:

r	25	28	35	45	60	100	150	166
d_{corr}	2.05	2.07	2.08	2.08	2.06	1.01	1.02	1.04

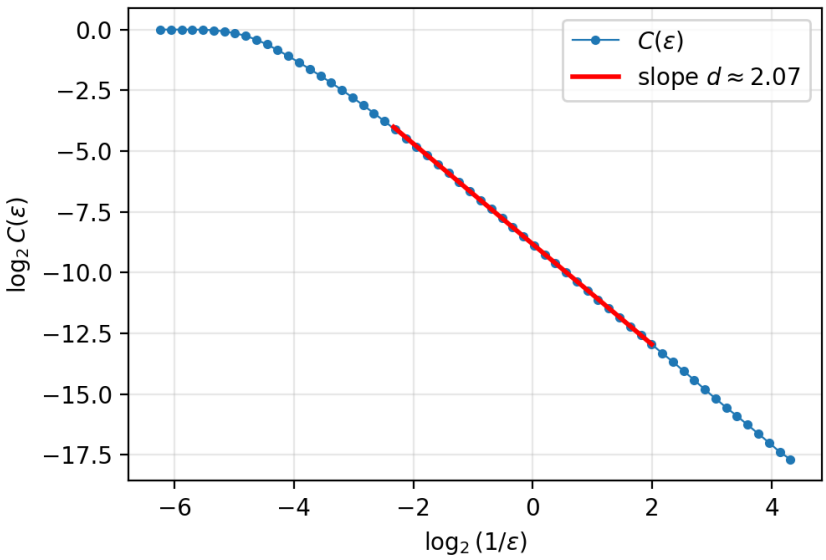


Figure 5: Correlation integral of the Lorenz attractor ($r = 28$, $\sigma = 10$, $b = 8/3$, $n = 15\,000$ points, $\Delta t = 0.25$).

Interpretation:

- For the chaotic values tested ($25 \le r \le 60$) the correlation dimension is just above 2, roughly 2.05–2.1, with very weak dependence on r. This is plausible because the volume contraction is very strong ($\nabla \cdot \mathbf{f} = -(\sigma + 1 + b) \approx -13.7$), so the attractor is nearly a surface (a Kaplan–Yorke estimate from the Lyapunov exponents at $r = 28$ gives about 2.06).
- For $r = 100$, 150, 166 the dimension is ≈ 1: these parameter values lie in periodic windows where the attractor is a stable limit cycle, a closed curve (dimension 1). For r below about 24 the long-time attractors are fixed points (dimension 0); for r above about 214 there is again a stable limit cycle.
- So the dimension is not a smooth function of r: it jumps between ≈ 2.05 (chaotic) and 1 (periodic), with alternating chaotic and periodic windows in between. Near the boundaries of the windows, long transients and intermittency make the numerical estimates unreliable. (At $r = 215$, where a stable cycle is expected, my program returned ≈ 1.5; I regard that value as an artifact and omit it, and I did not investigate its cause.)